Business Requirement

Inventory Management System (IMS) for Thrifted & Branded Surplus Clothing (with Item Approval Workflow)

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1.1 The System: Business Requirements

The Inventory Management System (IMS) is a web-based application that controls the full life cycle of thrifted and branded surplus clothing, from intake to sale, and prevents any garment from being sold before it has passed a physical inspection.

Thrifted and branded surplus garments vary widely in condition and authenticity, so every item must be physically examined by a Reviewer before it is offered for sale. The Reviewer checks garment quality, physical condition, tags, and authenticity, and then records the result in the system. The system does not perform the inspection. It records the decision, enforces the status rules that follow from it, and keeps an auditable history of every status change.

The business requirements are:

- Every new item must enter the system with the status “Pending” and must not be sellable until a Reviewer approves it.
- Only a physical inspection result recorded by a Reviewer can move an item from “Pending” to “In Stock”.
- A rejected item returns to “Pending” and must carry a mandatory rejection reason.
- Only “In Stock” items can be sold, and a sold item moves to “Sold”.
- Management must be able to search inventory, monitor stock levels, and generate status reports.

1.2 Functional Requirement Specifications (FRS)

Staff

- Log in with a role-based account and access only Staff functions.
- Add a new item with brand, size, condition, price, and photo; the system sets its status to “Pending” automatically.
- View the items they have added and the current status of each.
- Process a sale for an item whose status is “In Stock”.
- Be prevented by the system from selling any item that is “Pending” or “Sold”.

Reviewer

- Log in with a role-based account and access only Reviewer functions.
- View the Pending Items queue.
- Physically inspect each garment and record the result in the system.
- Select “Approve” so that the item status changes to “In Stock”.

- Select “Reject” and enter a mandatory rejection reason so that the item returns to “Pending” with the reason recorded.

Admin

- Log in with a role-based account and access only Admin functions.
- Search the inventory by brand, size, condition, status, and price.
- Monitor stock levels and view low-stock alerts.
- Generate status reports covering Pending, In Stock, and Sold items, including rejection reasons and status history.

System-wide rules

- The Status Guard Rule prevents any item from skipping a status; an item must be “In Stock” before it can be “Sold”.
- Every status change is logged with the item, the previous status, the new status, the user, and the date and time.
- Lifecycle: Pending → Approved (In Stock) or Rejected (back to Pending with reason) → Sold.

1.3 Non-Functional Requirement Specifications (NFRS)

Category	Requirement
Usability	Approving or rejecting an item must take no more than 2 clicks from the Pending Items queue.
Performance	Inventory search results must load in under 2 seconds.
Reliability	The system must achieve 99% uptime during the operating and testing period.
Security	Access is controlled by role-based access control (RBAC), and all passwords are stored as hashed values.
Scalability	The system must support 2,000 or more inventory items without loss of function or performance.

1.4 Automation Testing Tools

Postman collections are the automation tool for the IMS. The collections are grouped by module and run with the Postman Collection Runner against the test environment. They check that:

- Status rules are enforced: new items start as "Pending", Approve sets "In Stock", Reject returns the item to "Pending", and a "Pending" item cannot be sold.
- API constraints are applied: a rejection requires a reason, required fields and prices are validated, and each endpoint returns the correct status code and error message.
- Role-based access is applied: Staff, Reviewer, and Admin can call only the endpoints allowed for their role.
- Inventory search returns correct results in under 2 seconds.

Checks that cannot be done through the API, such as the 2-click approve/reject requirement and screen layout, are done manually.